Network Analysis - Blerd

AUTHOR
BDH

Data Set Up

```
# 1.
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3

── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   4.0.1     ✔ tibble    3.2.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.0.4     
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to b
```

```
library(qgraph)
library(bootnet)
```

```
This is bootnet 1.6
For questions and issues, please see github.com/SachaEpskamp/bootnet.
```

```
library(NetworkComparisonTest)
```

```
Warning: package 'NetworkComparisonTest' was built under R version 4.4.3
```

```
# 2.
setwd("/Users/ben/Desktop/Blerd/Data Analysis")

# 3. Load your data
datum = read.csv("Blerd.csv")
datumlong = read.csv("BlerdData_long.csv")

# 4. Ensure categorical variables are properly set
datum$SexF = as.factor (datum$Sex)
datum$TreatmentF = as.factor(datum$Treatment)
datum$TimeF = as.factor(datum$Time)
datum$IncubatorF = as.factor (datum$Incubator)
datum$SideF = as.factor (datum$Side)
datum$PostGlucose = as.numeric (datum$PostGlucose)
```

```
Warning: NAs introduced by coercion
```

```
datum$PreGlucose = as.numeric (datum$PreGlucose)
datum$PostKetones = as.numeric (datum$PostKetones)
```

```
Warning: NAs introduced by coercion
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```
datum$PreKetones = as.numeric (datum$PreKetones)
datum$PostDamage = as.numeric (datum$PostDamage)
datum$PreDamage = as.numeric (datum$PreDamage)
datum$PostCORT = as.numeric (datum$PostCORT)
datum$PreCORT = as.numeric (datum$PreCORT)
datum$AccMass = as.numeric (datum$AccMass)
datum$AvgMaxTemp = as.numeric (datum$AvgMaxTemp)

# 5. Subset data so there are no dead individuals included
datum1 = datum %>%
  filter(!Died %in% c("Died", "PIT.Death"))

# 6. Take out any of the blank cells that cause problems when graphing
datum2 = datum1 %>%
filter(PostGlucose != "", PostGlucose != "Dead",
         AvgMaxTemp != "", AvgMaxTemp != "Dead",
         PostKetones != "", PostKetones != "Dead",
         PostDamage != "", PostDamage != "Dead",
         PostCORT != "", PostCORT != "Dead")
datum2$TreatmentF = as.factor(datum2$Treatment)

datum_final = datum2
```

Correlation Matrix

```
# Define your variables
cor_vars = c("PostKetones", "PostGlucose", "PostCORT", "PostDamage", "AccMass",
             "AvgMaxTemp")

# Calculate correlation matrix
# 'use = "pairwise.complete.obs"' handles any missing values (NAs) row-by-row
cor_matrix <- cor(datum2[, cor_vars], use = "pairwise.complete.obs")

# Print the result rounded to 2 decimal places
round(cor_matrix, 2)
```

```
            PostKetones PostGlucose PostCORT PostDamage AccMass AvgMaxTemp
PostKetones        1.00        0.05     0.08       0.14    0.09       0.35
PostGlucose        0.05        1.00     0.16      -0.20   -0.21       0.48
PostCORT           0.08        0.16     1.00      -0.16   -0.14       0.30
PostDamage         0.14       -0.20    -0.16       1.00    0.03      -0.24
AccMass            0.09       -0.21    -0.14       0.03    1.00      -0.09
AvgMaxTemp         0.35        0.48     0.30      -0.24   -0.09       1.00
```

Network Correlation Analysis - Treatment Effects

```
# 1. Define variables that are wanted for network analysis
cor_vars = c("AvgMaxTemp", "PostKetones", "PostGlucose", "PostCORT", "PostDamage")

# 2. Estimate Networks
net_control = estimateNetwork(
  datum_final[datum_final$TreatmentF == "Control", cor_vars],
  default = "pcor",
  corMethod = "cor_auto"
  )
```

```
Estimating Network. Using package::function:
  - qgraph::qgraph(..., graph = 'pcor') for network computation
  - qgraph::cor_auto for correlation computation
    - using lavaan::lavCor
  - psych::corr.p for significance thresholding
```

```
net_heat = estimateNetwork(
  datum_final[datum_final$TreatmentF == "Heat", cor_vars],
  default = "pcor",
  corMethod = "cor_auto"
  )
```

```
Estimating Network. Using package::function:
  - qgraph::qgraph(..., graph = 'pcor') for network computation
  - qgraph::cor_auto for correlation computation
    - using lavaan::lavCor
  - psych::corr.p for significance thresholding
```

```
  #Example code:
    #estimateNetwork()
      #What this means: estimateNetwork() is the command from the bootnet package, and i
    #datum_final[datum_final$TreatmentF == "Control", cor_vars]
      #What this means: this is the data that we are interested in working with. In step
    #default = "pcor",
      #What this means: This identifies unique relationships between variables by contro
    #corMethod = "cor_auto"
      #What this means: This identifies what kind of data we have (continous or categori

# 3. Extract numbers for visualization
  #Step 2 calculated all of the partial correlations using statistical models, now we wa
weights_control = net_control$graph
weights_heat = net_heat$graph
weights_diff = weights_heat - weights_control

# 4. Calculate the global strength
  #Sum of all absolute edge weights (divided by 2 because the matrix is mirrored). The a
strength_control = sum(abs(weights_control)) / 2
strength_heat    = sum(abs(weights_heat)) / 2
  #Print the results
print(paste("Global Strength - Control:", round(strength_control, 3)))
```

```
[1] "Global Strength - Control: 1.516"
```

```
print(paste("Global Strength - Heat:", round(strength_heat, 3)))
```

```
[1] "Global Strength - Heat: 2.414"
```

```
# 5. Run a Network Comparison Test. In short, this gives you a p-value to see if the con
nct_treatment = NCT(
  datum_final[datum_final$TreatmentF == "Control", cor_vars],
  datum_final[datum_final$TreatmentF == "Heat", cor_vars],
  it = 1000,                 #Number of permutations
  test.edges = TRUE,         #Tests if specific connections changed
  test.centrality = TRUE    #Tests if specific nodes became more important
)
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```
  #Print results
summary(nct_treatment)
```

```
 INDEPENDENT GROUPS GAUSSIAN NETWORK COMPARISON TEST 

 P-VALUE CORRECTION: none 

 NETWORK INVARIANCE TEST 
 Test statistic M: 0.01587044 
 p-value 0.08491508 

 GLOBAL STRENGTH INVARIANCE TEST 
 Global strength per group:  0 0.01587044 
 Test statistic S:  0.01587044 
 p-value 0.08491508 

 EDGE INVARIANCE TEST 
          Var1        Var2    p-value Test statistic E
6   AvgMaxTemp PostKetones 0.02997003       0.01587044
11  AvgMaxTemp PostGlucose 1.00000000       0.00000000
12 PostKetones PostGlucose 1.00000000       0.00000000
16  AvgMaxTemp    PostCORT 1.00000000       0.00000000
17 PostKetones    PostCORT 1.00000000       0.00000000
18 PostGlucose    PostCORT 1.00000000       0.00000000
21  AvgMaxTemp  PostDamage 1.00000000       0.00000000
22 PostKetones  PostDamage 1.00000000       0.00000000
23 PostGlucose  PostDamage 1.00000000       0.00000000
24    PostCORT  PostDamage 1.00000000       0.00000000


 CENTRALITY INVARIANCE TEST 
 Nodes tested: AvgMaxTemp PostKetones PostGlucose PostCORT PostDamage 
 Centralities tested: strength expectedInfluence 
 Test statistics C: 
               strength expectedInfluence
AvgMaxTemp  -0.01587044       -0.01587044
PostKetones -0.01587044       -0.01587044
PostGlucose  0.00000000        0.00000000
PostCORT     0.00000000        0.00000000
PostDamage   0.00000000        0.00000000

 p-values: 
              strength expectedInfluence
AvgMaxTemp  0.08391608        0.08391608
PostKetones 0.03096903        0.03096903
PostGlucose 1.00000000        1.00000000
PostCORT    1.00000000        1.00000000
PostDamage  1.00000000        1.00000000
```

```
  #p=0.086
```

Network Correlation - Sex differences in heat

```
# 1. Define variables that are wanted for network analysis
cor_vars = c("AvgMaxTemp", "PostKetones", "PostGlucose", "PostCORT", "PostDamage")
  #Create datasets for heated males and females
    #Male
male_data_final = datum_final %>%
  filter(TreatmentF == "Heat", SexF == "M") %>%
  dplyr::select(all_of(cor_vars)) %>%
  drop_na()
    #Female
female_data_final = datum_final %>%
  filter(TreatmentF == "Heat", SexF == "F") %>%
  dplyr::select(all_of(cor_vars)) %>%
  drop_na()


# 2. Estimate Networks
net_heat_male = estimateNetwork(
  datum_final[datum_final$TreatmentF == "Heat" & datum_final$SexF == "M", cor_vars],
  default = "cor", corMethod = "cor_auto"
)
```

```
Estimating Network. Using package::function:
  - qgraph::cor_auto for correlation computation
    - using lavaan::lavCor
  - psych::corr.p for significance thresholding
```

```
net_heat_female = estimateNetwork(
  datum_final[datum_final$TreatmentF == "Heat" & datum_final$SexF == "F", cor_vars],
  default = "cor", corMethod = "cor_auto"
)
```

```
Estimating Network. Using package::function:
  - qgraph::cor_auto for correlation computation
    - using lavaan::lavCor
  - psych::corr.p for significance thresholding
```

```
  #NOTE: we have changed this to be just correlational, so this differs from the above c

# 3. Extract numbers for visualization
  #Step 2 calculated all of the partial correlations using statistical models, now we wa
weights_male = net_heat_male$graph
weights_female = net_heat_female$graph

# 4. Calculate the global strength
  #Sum of all absolute edge weights (divided by 2 because the matrix is mirrored). The a
strength_male = sum(abs(weights_male)) / 2
strength_female = sum(abs(weights_female)) / 2
  #Print the results
print(paste("Global Strength - Male:", round(strength_male, 3)))
```

```
[1] "Global Strength - Male: 3.262"
```

```
print(paste("Global Strength - Female:", round(strength_female, 3)))
```

```
[1] "Global Strength - Female: 2.325"
```

```
# 5. Run a Network Comparison Test. In short, this gives you a p-value to see if the hea
nct_sex_diff = NCT(
  datum_final[datum_final$TreatmentF == "Heat" & datum_final$SexF == "M", cor_vars],
  datum_final[datum_final$TreatmentF == "Heat" & datum_final$SexF == "F", cor_vars],
  it = 1000,
  estimator = function(x){ qgraph::cor_auto(x) }, #This line calculates just correlation
  test.edges = TRUE,
  test.centrality = TRUE
)
```

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```
# See the results
summary(nct_sex_diff)
```

```
 INDEPENDENT GROUPS GAUSSIAN NETWORK COMPARISON TEST 

 P-VALUE CORRECTION: none 

 NETWORK INVARIANCE TEST
 Test statistic M: 0.930634 
 p-value 0.1738262 

 GLOBAL STRENGTH INVARIANCE TEST
 Global strength per group:  3.261635 2.324556 
 Test statistic S:  0.9370786 
 p-value 0.2177822 

 EDGE INVARIANCE TEST 
          Var1        Var2    p-value Test statistic E
6   AvgMaxTemp PostKetones 0.45054945       0.19794131
11  AvgMaxTemp PostGlucose 0.56743257       0.19873041
12 PostKetones PostGlucose 0.76923077       0.09403329
16  AvgMaxTemp    PostCORT 0.85514486       0.14378300
17 PostKetones    PostCORT 0.99000999       0.00630337
18 PostGlucose    PostCORT 0.20979021       0.50750856
21  AvgMaxTemp  PostDamage 0.11688312       0.60593441
22 PostKetones  PostDamage 0.09390609       0.93063405
23 PostGlucose  PostDamage 0.15884116       0.72080363
24    PostCORT  PostDamage 0.03596404       0.56398737


 CENTRALITY INVARIANCE TEST 
 Nodes tested: AvgMaxTemp PostKetones PostGlucose PostCORT PostDamage 
 Centralities tested: strength expectedInfluence 
 Test statistics C: 
              strength expectedInfluence
AvgMaxTemp   0.4593759       -0.46294050
PostKetones -0.3710074       -0.63235608
PostGlucose  0.1899004       -0.31799220
PostCORT     0.9553822        0.09360756
PostDamage   0.6405061       -2.82135945

 p-values: 
              strength expectedInfluence
AvgMaxTemp  0.33266733       0.711288711
PostKetones 0.47452547       0.363636364
PostGlucose 0.53846154       0.720279720
PostCORT    0.08791209       0.916083916
PostDamage  0.19880120       0.000999001
```

```
  #Network Invariance p=0.18
  #Global Strength Invariance p=0.24
```

Manuscript Graph - Treatment Network

```
# 1. Visualization Parameters
  #Create a fixed layout for the network graph. This locks the positions based on the He
L = qgraph(weights_heat, DoNotPlot = TRUE)$layout
  #Renaming the variables so they are more intitive for the reader
my_node_labels = c("Body\nTemp", "Ketones", "Glucose", "CORT", "DNA\nDamage")
  #We define a list of common arguments to ensure consistency across all three network p

common_args = list(
  layout = L,
  labels = my_node_labels,
  label.cex = 1,
  label.font = 2,
  edge.labels = TRUE,
  edge.label.cex = 2.5,
  edge.label.font = 2,
  edge.label.color = "black",
  edge.label.bg = "white",
  edge.label.margin = 0.01,
  posCol = "#084594",
  negCol = "firebrick4",
  vsize = 20,
  curveAll = TRUE,
  curveDefault = 0.5,
  fade = TRUE,
  border.width = 1
)

# 2. Organizing the .PNG file. NOTE: in order for this code to run you have to run it al
  #title of file
path = "Network_Treatment.png"
  # Open a high-resolution PNG file (300 DPI) with width and height values
png(path, width = 7200, height = 3000, res = 300, bg = "transparent")
  #This creates a space at the top of the whole file so that there is room for titles
par(mfrow = c(1, 3), oma = c(0, 0, 15, 0), mar = c(3, 3, 3, 3))

# 3. Making the figure
  #Panel A - Control Network
do.call(qgraph, c(list(input = weights_control), common_args))
mtext("A: Control Network", side = 3, line = 5, cex = 3, font = 2, outer = TRUE, adj = 0
  # Panel B - Heat Network
do.call(qgraph, c(list(input = weights_heat), common_args))
mtext("B: Heat Network", side = 3, line = 5, cex = 3, font = 2, outer = TRUE, adj = 0.41
  # Panel C - Difference Network
do.call(qgraph, c(list(input = weights_diff), common_args))
mtext("C: Difference Network", side = 3, line = 5, cex = 3, font = 2, outer = TRUE, adj
  #This closes the PNG that we opened at the top of this chuck of code
dev.off()
```

```
quartz_off_screen 
                2 
```

```
system(paste("open", shQuote(path)))
par(mfrow = c(1, 1))
```